Differential Nocuity: How Ambiguity Type and Model Architecture Shape Safety Violations in LLM Agents

Anonymous ACL submission

Abstract

Large Language Model (LLM) agents increasingly execute real-world tasks governed by natural-language policies, yet these policies inevitably contain ambiguities. We present the first causal, type-granularity, cross-model study of how policy ambiguity drives safety violations in LLM agents. Using a matched-pair design with 300 clause pairs spanning six ambiguity types and a two-layer taxonomy (specification-layer vs. linguistic-layer), we evaluate five architecturally diverse models—GPT-5.4, GPT-4.1, Claude Sonnet 4.6, Qwen3-235B, and DeepSeek-V3—across 2,997 agentic episodes in the τ^2-bench framework. We find that policy ambiguity is *universally nocuous*: ambiguous clauses increase safety violations by $+34.0$ pp (OR=15.1, $p < 10^{-50}$), an effect replicated across all five models and confirmed as ambiguity-specific by a perturbation control ($\Delta_{\text{perturb}} = 3.33$ pp, n.s.). However, not all ambiguity is equally dangerous. A three-tier type hierarchy emerges—incompleteness (53.6%) $\gg$ mid-cluster (36.8–42.8%) $\gg$ coreferential (31.6%)—and a significant model$\times$type interaction reveals that authorization-scope vulnerability ranges 52 pp across models (12% Claude to 64% GPT-5.4), while incompleteness remains universally dangerous (46–62%). Error analysis of 450 violation traces uncovers type-diagnostic failure signatures and pervasive *Ambiguity Blindness*: agents show $<$5.2% awareness of ambiguity, with $>$91% surface adoption of a single interpretation. Our *Differential Nocuity* framework connects Berry & Kamsties' nocuous-ambiguity theory to modern safety alignment, demonstrating that safety training differentially mitigates specific ambiguity types while leaving others as universal blind spots.

1 Introduction

Consider an airline customer service agent operating under the policy: *"Items may be returned within 30 days."* The scopal ambiguity of *"items"*—does it scope over all categories, including non-refundable custom orders?—leads the agent to approve a full refund for a customized product. The unambiguous variant, *"Standard items only may be returned within 30 days,"* triggers a correct refusal. One word of ambiguity; a $200 unauthorized refund.

LLM-based agents are rapidly deployed for customer service, workflow automation, and enterprise decision-making, governed by natural-language policy specifications (**??**). These policies are written by humans and inevitably contain ambiguities—a well-documented phenomenon in requirements engineering (RE) (**??**) and increasingly recognized in NLP (**?**). While recent benchmarks evaluate agent safety (**??**) and policy compliance (**?**), none isolate ambiguity as a causal variable or differentiate its types.

Berry & Kamsties (**?**) distinguish *nocuous* ambiguity (divergent interpretations causing downstream harm) from *innocuous* ambiguity (convergent interpretations). In human RE, shared organizational context renders most ambiguity innocuous. LLM agents, however, lack such context: they process policy text literally, without access to institutional knowledge, legislative intent, or the pragmatic norms that guide human interpretation. **Are policy ambiguities nocuous for LLM agents? Does the type of ambiguity matter? Does it depend on the model?**

We address these questions with a matched-pair causal study spanning five architecturally diverse models (three proprietary, two open-source), six ambiguity types organized in a two-layer taxonomy, and 2,997 agentic episodes in τ^2-bench (**?**). A perturbation control experiment confirms that the observed effect is ambiguity-specific, not attributable to general prompt brittleness. Our contributions are:

1. **Causal quantification.** Policy ambiguity increases safety violations by $+34.0$ pp (OR=15.1) across all five models—the strongest causal evidence of ambiguity's safety impact in the agent safety literature, backed by matched-pair design and perturbation control ($\Delta_{\text{perturb}} = 3.33$ pp, n.s.).

2. **Differential Nocuity framework.** We introduce *Differential Nocuity*: not all ambiguity types are equally dangerous, and vulnerability profiles differ across models. A three-tier type hierarchy places incompleteness (53.6%) far above coreferential (31.6%), while a model×type interaction ($\chi^2 = 24.01$, $p = 7.9 \times 10^{-5}$) shows authorization-scope vulnerability varying 52 pp across models—a canary for safety-training quality.

3. **Type-specific failure taxonomy and Ambiguity Blindness.** Analysis of 450 violation traces reveals type-diagnostic failure signatures (5/6 types share the same dominant mode across all five models) and quantifies *Ambiguity Blindness*: agents exhibit $<5.2\%$ meta-cognitive awareness of ambiguity, with $>91\%$ surface adoption.

2 Related Work

Ambiguity in NLP and Requirements Engineering. Ambiguity classification has a long history in RE (**????**) and computational linguistics (**??**). Li et al. (**?**) propose an 11-type NLP taxonomy; Massey et al. (**?**) identify six RE-specific types. Chantree et al. (**?**) pioneer computational nocuous-ambiguity detection. Our six-type, two-layer taxonomy synthesizes these traditions, crosswalking to both Li et al.'s NLP types and Massey et al.'s RE types.

Agent Safety Evaluation. Recent work evaluates agent safety through diverse lenses: ToolEmu (**?**) simulates tool-use risks; R-Judge (**?**) benchmarks safety reasoning; InjecAgent (**?**) studies prompt injection attacks. PhantomPolicy uses policy-violation scenarios but does not isolate ambiguity type. DRIP-R (**?**) classifies five RE ambiguity types and measures interpretation divergence across models, but does not link divergence to safety violations or provide causal evidence. Our work closes this gap: we causally quantify the safety impact of typed ambiguity across five model families.

Prompt Sensitivity and Robustness. LLM outputs are sensitive to superficial prompt variations (**??**). Our perturbation control distinguishes ambiguity-specific harm from general brittleness—non-ambiguity surface changes (synonym substitution, clause reordering) produce negligible effect ($+3.33$ pp, n.s.) compared to ambiguity ($+36.5$ pp).

Psycholinguistic Foundations. Good-enough processing (**?**) suggests humans sometimes adopt shallow parses of ambiguous sentences. LLM agents exhibit an analogous phenomenon—*Ambiguity Blindness*—adopting surface interpretations without meta-cognitive awareness. Our Disambiguation Channel Framework connects to the pragmatics-vs.-semantics distinction and expectation-based processing (**??**).

3 Ambiguity Taxonomy

We organize policy ambiguity into a two-layer, six-type taxonomy grounded in both NLP (**?**) and RE (**?**) traditions.

Specification Layer. These ambiguities arise from gaps or vagueness in the policy's *intent specification*, irresolvable without domain knowledge or stakeholder clarification:

- **Incompleteness**: Missing conditions, exceptions, or boundary cases (*e.g.*, "Refunds are available for defective products" without specifying a time limit).

- **Authorization Scope**: Vague permission or role boundaries (*e.g.*, "Authorized personnel may access the system" without specifying which functions).

- **Conditional Precedence (CP)**: Conflicting or underspecified rule ordering (*e.g.*, overlapping discount and return policies without priority).

Linguistic Layer. These ambiguities arise from natural-language structure and can sometimes be resolved through context, tool responses, or world knowledge:

- **Scopal**: Quantifier, negation, or modal scope interactions (*e.g.*, "All customers are not eligible" admits universal-negation and negation-of-universal readings).

- **Lexical**: Polysemy or vague terms (*e.g.*, "reasonable" timeframe).

- **Coreferential**: Ambiguous pronoun or relative-clause attachment (*e.g.*, "The agent should contact the customer who filed the complaint and their manager").

The specification–linguistic distinction maps onto *disambiguation channel availability*: linguistic ambiguities have partial external resolution channels (tool outputs, conversational context), while specification ambiguities force the agent to guess.

4 Method

4.1 Experimental Design

We employ a matched-pair within-subject design. For each of 300 policy clauses (50 per type), we construct an *ambiguous* variant and a minimally-edited *unambiguous* variant. Each clause pair is evaluated under both conditions across all five models, yielding $300 \times 2 \times 5 = 3{,}000$ planned episodes (2,997 completed; 99.7% completion rate).

4.2 Clause Generation and Validation

We generated 300 clause pairs (50 per type $\times$ 6 types) following Chantree et al.'s (**?**) ambiguity-annotation standards. Conditional-precedence clauses used a mixed format (60% labeled, 40% naturalistic rule rewrites) to avoid format artifacts. Quality validation achieved a 98.6% pass rate (296/300 first-pass; 4 revised), with inter-annotator agreement $\kappa = 0.92$.

Table 1 summarizes clause properties. Specification-layer clauses require substantially more tokens to disambiguate (mean $\Delta = 28.3$ tokens) than linguistic-layer clauses ($\Delta = 7.3$; $t = 14.2$, $p < 0.001$), operationalizing the distinction in disambiguation channel availability.

4.3 Realism Validation

We compared the distributional properties of our 300 synthetic clauses against 147 naturally-occurring ambiguous clauses from the DRIP-R corpus (**?**) across four linguistic dimensions. Kolmogorov–Smirnov tests revealed no significant distributional differences on any dimension (Table 2), confirming ecological validity.

Type	Layer	n	Amb	Unamb	Δ	Pass
Incompleteness	Spec	50	48.2	82.8	34.6	50/50
Auth. scope	Spec	50	44.6	80.7	36.1	48/50
Cond. prec.	Spec	50	38.1	52.3	14.2	50/50
Scopal	Ling	50	36.4	45.1	8.7	50/50
Lexical	Ling	50	41.7	47.0	5.3	49/50
Coreferential	Ling	50	44.8	52.6	7.8	49/50

Table 1: Clause distribution. **Amb/Unamb**: mean token length of ambiguous/unambiguous variants. **Δ**: token delta (disambiguation cost). **Pass**: clauses passing quality validation on first pass.

Property	Synth.	DRIP-R	KS	p
Sent. length (tok)	24.3$\pm$8.1	22.8$\pm$7.6	.087	.23
Parse depth	5.7$\pm$1.9	5.4$\pm$1.7	.094	.18
Flesch–Kincaid	12.8$\pm$2.4	13.2$\pm$2.6	.072	.41
Type–token ratio	0.71$\pm$0.08	0.68$\pm$0.09	.112	.09

Table 2: Realism validation: synthetic clauses vs. DRIP-R natural ambiguities. No significant distributional differences ($p > 0.05$ on all KS tests).

4.4 Agent Platform and Models

Episodes are executed in τ^2-bench (**?**), a sandboxed agentic evaluation framework spanning airline and retail customer-service domains. We evaluate five models spanning three providers and two architecture families:

- **Proprietary dense**: GPT-5.4, GPT-4.1, Claude Sonnet 4.6

- **Open-source MoE**: Qwen3-235B

- **Open-source dense**: DeepSeek-V3

4.5 Evaluation: Cross-Judge Design

Each episode is scored for policy violations using an LLM-as-judge paradigm with a cross-judge design to avoid self-evaluation bias: GPT-5.4 episodes are judged by GPT-4.1; all other models' episodes are judged by GPT-5.4. Cross-judge agreement is high ($\kappa = 0.91$).

Judge Calibration. We calibrated the automated judge against human ground truth on a stratified sample of 100 episodes annotated by two human raters (Table 3). Human inter-annotator agreement was $\kappa = 0.87$ (93% raw agreement). The automated judge achieved $F_1 = 0.90$ overall (precision 0.91, recall 0.89), with highest performance on clear violations ($F_1 = 0.93$) and adequate performance on borderline cases ($F_1 = 0.80$). Cross-judge asymmetry was 3.1 pp, well

Severity tier	N	Prec.	Rec.
Clear violation	38	0.94	0.92
Borderline	24	0.78	0.82
No violation	38	0.93	0.91
Overall	**100**	**0.91**	**0.89**

Table 3: Judge calibration against human ground truth ($F_1 = 0.90$; human $\kappa = 0.87$).

below the 10 pp threshold, with a slight over-detection bias (+2.3 pp FPR).

4.6 Reliability: Test–Retest Analysis

To quantify measurement stability, we conducted a test–retest study: 30 clause pairs × 3 independent replicates × 2 conditions × 2 models (GPT-5.4, GPT-4.1) = 360 episodes. The overall intraclass correlation coefficient (ICC) was 0.919 [0.892, 0.940] (excellent), confirming that per-clause violation rates are highly reproducible. Binary ICC was 0.847 [0.803, 0.883] (good). Per-type ICCs revealed a layer-stratified pattern: linguistic-layer types achieved uniformly excellent ICCs (scopal 0.989, lexical 0.981, coreferential 0.976), while specification-layer types achieved good ICCs (authorization scope 0.857, CP 0.856, incompleteness 0.807). This pattern provides convergent evidence for the Disambiguation Channel Framework (§7): specification-layer ambiguities, lacking external resolution channels, produce more stochastic agent behavior.

5 Results

5.1 C1: Binary Effect—Ambiguity Is Universally Nocuous

Across all 2,997 episodes, ambiguous policy clauses produced a violation rate of 40.9%, compared to 2.9% for matched unambiguous variants—a causal effect of +34.0 pp (bootstrap 95% CI: [31.2, 36.9]). The pooled McNemar odds ratio was 15.1 [10.78, 21.17], and the effect was replicated across all five models without exception (Table 4). Logistic regression with condition, model, type, and token delta as covariates confirmed ambiguity as the dominant predictor ($\beta_{\text{condition}} = 2.278$, $z = 19.65$, $p < 0.001$).

Perturbation Control. To distinguish ambiguity-specific harm from general prompt brittleness, we applied non-ambiguity perturbations (synonym substitution and clause reordering) to 30 unambiguous clause pairs

Model	N	Amb	Unamb	Δ	OR	p
DeepSeek-V3	600	49.7%	0.7%	+39.0	22.8	<1e-27
GPT-5.4	600	41.0%	2.7%	+38.3	25.1	<1e-25
GPT-4.1	600	40.7%	6.0%	+34.7	10.8	<1e-20
Qwen3-235B	600	43.1%	0.7%	+32.4	17.3	<1e-22
Claude-S-4.6	600	30.3%	4.6%	+25.7	9.0	<1e-17
Pooled	**2997**	**40.9%**	**2.9%**	**+34.0**	**15.1**	**<1e-50**

Table 4: C1: Per-model binary effect. **Amb/Unamb**: violation rates under ambiguous/unambiguous conditions. **Δ**: effect size in percentage points. All models show large, highly significant effects.

Condition	N	Viol.%	Δ	95% CI
Original (unamb.)	60	4.2%	—	—
Synonym subst.	30	8.3%	+4.2	[−3.1, 11.5]
Clause reorder	30	6.7%	+2.5	[−4.6, 9.6]
All perturbations	60	7.5%	+3.33	[−1.8, 8.5]
Ambiguity (ref.)	—	—	+36.5	[33.0, 40.0]

Table 5: Perturbation control. Non-ambiguity surface changes produce negligible, non-significant effects; ambiguity produces massive effect.

across 2 models (120 episodes). Perturbations produced a negligible $\Delta_{\text{perturb}} = 3.33$ pp (95% CI: [−1.8, 8.5], containing zero; $p = 0.20$), compared to the ambiguity effect of +36.5 pp (Table 5). The ratio $\Delta_{\text{perturb}}/\Delta_{\text{ambiguity}} = 0.091$, confirming that >90% of the observed effect is ambiguity-specific.

5.2 C2: Type Hierarchy—Not All Ambiguity Is Equal

The six ambiguity types differ significantly in violation severity (omnibus $\chi^2 = 28.02$, $p = 3.6 \times 10^{-5}$, $V = 0.137$; power = 0.993, MDE = 0.093). A three-tier structure emerges from TOST equivalence testing (Table 6):

- **High**: Incompleteness (53.6%)—universally dangerous across all models.
- **Mid**: Lexical (42.8%), authorization scope (41.6%), scopal (39.4%), CP (36.8%)—pairwise equivalent at ±15 pp (8/15 pairs).
- **Low**: Coreferential (31.6%)—partial disambiguation via tool responses.

The incompleteness–coreferential gap is 22.0 pp. At ±10 pp, only 4/15 type pairs are equivalent, confirming that ambiguity types are not interchangeable.

Type	Layer	N_{amb}	Viol.%	Tier
Incompleteness	Spec	250	53.6%	HIGH
Lexical	Ling	250	42.8%	MID
Auth. scope	Spec	250	41.6%	MID
Scopal	Ling	250	39.4%	MID
Cond. prec.	Spec	250	36.8%	MID
Coreferential	Ling	250	31.6%	LOW

Table 6: C2: Type hierarchy (ambiguous condition, 5 models pooled). Three-tier structure from TOST equivalence analysis.

5.3 C3: Layer Effect—Specification > Linguistic

Specification-layer ambiguities produce higher violation rates on average: 44.0% vs. 37.9% for linguistic-layer ($\Delta = 6.08$ pp, $\chi^2 = 5.73$, $p = 0.017$, $V = 0.062$, OR $= 1.29$ [1.05, 1.58]). Within-judge subgroup analysis confirms significance in both groups: Group 1 (judged by GPT-4.1: GPT-5.4 + DeepSeek) shows $\Delta = 9.5$ pp ($p = 0.021$); Group 2 (judged by GPT-5.4: GPT-4.1 + Claude + Qwen3) shows $\Delta = 6.8$ pp ($p = 0.038$).

However, the aggregate masks substantial model-dependence. Per-model layer gaps range from $+14.0$ pp (GPT-5.4) to -0.9 pp (Qwen3), driven primarily by authorization-scope rates: models with high auth-scope vulnerability (GPT-5.4: 64%) inflate the specification-layer average, while models with low auth-scope vulnerability (Claude: 12%) show negligible layer gaps. This model-dependence motivates the interaction analysis in §5.4.

Token delta partially mediates the layer effect: each additional disambiguation token increases violation odds (logistic $\beta = 0.022$, $z = 3.49$, $p = 0.0005$), consistent with the Disambiguation Channel Framework.

5.4 C4: Model × Type Interaction—The Star Finding

Models show significantly different overall vulnerability levels ($\chi^2 = 31.02$, $p = 3.0 \times 10^{-6}$, $V = 0.102$), with Claude (30.3%) the most robust and DeepSeek (49.7%) the most vulnerable. Critically, the model×type interaction is highly significant in the ambiguous condition ($\chi^2 = 24.01$, $p = 7.9 \times 10^{-5}$): models do not merely differ in *level* but in *profile* (Table 7).

Three interaction patterns emerge:

Type	GPT-5.4	GPT-4.1	Claude	Qwen3	DeepSk.	Range
Auth. scope	64%	46%	12%	26%	60%	52 pp
Coreferential	18%	32%	22%	42%	44%	26 pp
Lexical	42%	34%	32%	52%	54%	22 pp
Incompleteness	46%	58%	48%	62%	54%	16 pp
Scopal	42%	38%	32%	37%	48%	16 pp
Cond. prec.	34%	36%	36%	40%	38%	6 pp

Table 7: C4: Model×type interaction (ambiguous condition, violation %). Authorization scope shows the widest cross-model range (52 pp); conditional precedence is most model-invariant (6 pp).

Model	χ^2	p	V	Profile
GPT-5.4	23.44	.0003	.28	Selective
Claude	17.87	.003	.24	Selective
Qwen3	15.89	.007	.23	Moderate
GPT-4.1	9.89	.078	.18	Borderline
DeepSeek	6.31	.277	.14	Uniform

Table 8: C4: Within-model type differentiation. Models range from strongly selective (GPT-5.4, Claude) to uniformly vulnerable (DeepSeek).

Authorization scope as canary type. The 52 pp cross-model range (Claude 12% to GPT-5.4 64%) makes auth-scope the type most sensitive to safety-training quality. Claude's Reinforcement Learning from Human Feedback (RLHF) appears to specifically train authorization-boundary awareness, while DeepSeek (60%) and GPT-5.4 (64%) lack this targeted protection.

Incompleteness as universal blind spot. All five models show high vulnerability (46–62%, range 16 pp). Safety training cannot compensate when the policy simply omits necessary information.

Conditional precedence as model-invariant. The narrowest range (6 pp, 34–40%) suggests explicit rule conflicts are handled similarly across architectures.

Per-model within-type χ^2 tests (Table 8) reveal a gradient of type differentiation: GPT-5.4 ($V = 0.28$, $p = 0.0003$) and Claude ($V = 0.24$, $p = 0.003$) show strong differentiation; Qwen3 ($V = 0.23$, $p = 0.007$) shows moderate differentiation; GPT-4.1 ($V = 0.18$, $p = 0.078$) is borderline; and DeepSeek ($V = 0.14$, $p = 0.277$) shows no significant differentiation—uniformly vulnerable across all types.

Fleiss $\kappa = 0.384$ (fair) across the five models

confirms that inter-model agreement on *which specific clauses* trigger violations is only moderate—the binary effect is universal, but the clause-level question is model-dependent.

6 Error Analysis

We analyzed 450 violation traces (90 per model, stratified across all six types) using an automated four-category failure-mode taxonomy, validated by Gwet's AC1 $= 0.73$ (80% raw agreement; Cohen's $\kappa = 0.594$, deflated by the κ paradox in high-agreement/low-diversity categories).

6.1 C5: Type-Diagnostic Failure Signatures

Five of six types share the same dominant failure mode across all five models (Table 9), a pattern highly unlikely by chance (binomial $p < 0.001$ per type). Failure modes are type-determined, not model-determined:

- **Auth. scope** $\rightarrow$ unauthorized escalation (76%): agents systematically exceed permission boundaries.
- **Incompleteness** $\rightarrow$ assumption-based action (63%): agents impute missing conditions and act ("guess-and-act").
- **Scopal** $\rightarrow$ scope misapplication (74%): agents adopt the wrong scope reading.
- **Lexical** $\rightarrow$ assumption-based action (65%): agents treat word ambiguity as an information gap, not a misparse.
- **CP** $\rightarrow$ scope misapplication (38%) + arbitrary rule selection (28%): mixed, reflecting genuine rule conflicts.
- **Coreferential** $\rightarrow$ no dominant mode (flat: 31% scope, 29% assumption, 18% escalation): diverse resolution strategies enabled by partial disambiguation from tool responses.

A layer-level pattern emerges: linguistic-layer types are dominated by scope misapplication (39% ling avg vs. 15% spec avg), reflecting surface misparse. Specification-layer types are dominated by unauthorized escalation and assumption-based action combined (72%), reflecting active guessing under information absence.

Type	Unauth.	Assump.	Scope	Arb.
Auth. scope	76%	12%	7%	5%
Incompleteness	18%	63%	11%	8%
Scopal	8%	14%	74%	4%
Lexical	6%	65%	22%	7%
Cond. prec.	16%	18%	38%	28%
Coreferential	18%	29%	31%	22%

Table 9: Failure mode distribution by type (5-model pooled, %). **Unauth.**: unauthorized escalation. **Assump.**: assumption-based action. **Scope**: scope misapplication. **Arb.**: arbitrary rule selection. Each type has a diagnostic dominant mode except coreferential.

Model	Aware	Adopt	Escal.	Refuse
GPT-5.4	4.1%	92.4%	0.3%	2.1%
GPT-4.1	10.7%	85.6%	1.6%	3.2%
Claude	6.3%	89.2%	0.8%	4.7%
Qwen3	3.8%	93.1%	0.2%	1.8%
DeepSeek	2.9%	95.2%	0.1%	1.4%
Pooled	**5.2%**	**91.3%**	**0.7%**	**2.6%**

Table 10: Ambiguity Blindness across five models. **Aware**: traces showing meta-cognitive awareness. **Adopt**: surface interpretation adopted. **Escal.**: escalated to user. **Refuse**: conservative refusal.

6.2 Ambiguity Blindness

Across all 450 traces, agents exhibited meta-cognitive awareness of ambiguity in only 5.2% of violation episodes (Table 10). In 91.3% of cases, agents adopted a surface interpretation without hesitation; only 0.7% escalated to the user. Awareness was inversely correlated with action-orientation: frontier models (GPT-5.4: 4.1%, DeepSeek: 2.9%) showed *lower* awareness than the mid-tier GPT-4.1 (10.7%), suggesting that more capable, action-oriented models are less likely to pause before acting.

7 Discussion

Differential Nocuity. Berry & Kamsties (**?**) frame nocuity as inter-reader divergence. Our results extend this framework: all policy ambiguity is nocuous for LLM agents ($+34.0$ pp), but *how* it is nocuous varies by type and model. We propose a third category—*Systematic Nocuity*—where all models converge on the same wrong interpretation (27% of ambiguous episodes). This is more dangerous than divergent nocuity: it is invisible in inter-model agreement metrics and appears "correct" until checked against ground truth. The remaining violations split between divergent nocuity (42%, different wrong readings) and innocuous

outcomes (31%, correct despite ambiguity).

Disambiguation Channel Framework. The specification–linguistic layer distinction reflects disambiguation channel availability. Linguistic ambiguities (coreferential, lexical) can sometimes be resolved through tool outputs, conversational context, or world knowledge. Specification ambiguities (auth-scope, incompleteness) have no external resolution channel: the information is simply absent from the policy, forcing the agent to guess. Three independent lines of evidence support this framework: (1) the aggregate layer effect ($\Delta = 6.08$ pp, $p = 0.017$); (2) per-type ICC stratification (ling-layer ICC 0.976–0.989 $>$ spec-layer 0.807–0.857), showing that agents behave more deterministically when disambiguation channels exist; and (3) failure mode distributions (ling $\rightarrow$ scope misapplication; spec $\rightarrow$ guess-and-act). Coreferential ambiguity, with its elevated conservative-refusal rate (8.5% vs. 2.5% average), serves as a boundary case: tool responses provide partial disambiguation cues, enabling more diverse resolution strategies.

Safety Training Selectivity. The model$\times$type interaction reveals that safety training differentially addresses specific ambiguity types. Claude's remarkable auth-scope robustness (12% vs. 52% five-model average) suggests its RLHF specifically addresses authorization-boundary reasoning. DeepSeek's uniformly high vulnerability ($V = 0.14$, n.s.) suggests less type-specific safety alignment. Incompleteness remains a universal blind spot: no amount of safety training compensates for literally absent policy information.

Practical Implications. Our findings recommend *both* type-aware policy auditing (prioritize incompleteness and authorization-scope clauses) *and* model-aware risk assessment (vulnerability profiles are model-specific). The low awareness rate ($<$5.2%) implies that ambiguity detection must be externalized—agents cannot be relied upon to flag their own uncertainty.

Limitations. Our study covers two domains (airline, retail); generalization to other policy domains is future work. Synthetic ambiguity injection, while necessary for causal inference and validated against DRIP-R distributions, may not capture all naturally-occurring ambiguity patterns. The automated judge, despite $F_1 = 0.90$ calibration, introduces measurement noise; a slight over-detection bias ($+2.3$ pp FPR) uniformly inflates all violation rates without affecting comparative conclusions. The C3 layer effect ($V = 0.062$) is small and model-dependent; the model$\times$type interaction (C4) is more theoretically fundamental.

8 Conclusion

We presented the first causal, cross-model study of policy ambiguity's impact on LLM agent safety. Policy ambiguity is universally nocuous ($+34.0$ pp across five models), but its nocuity is *differential*: incompleteness is universally dangerous (53.6%), authorization scope is a canary for safety-training quality (12–64% cross-model range), and coreferential ambiguity is partially mitigated by external disambiguation channels (31.6%). Agents are pervasively blind to their own ambiguity ($<$5.2% awareness), making external ambiguity detection essential. Our Differential Nocuity framework connects classical nocuous-ambiguity theory to modern safety alignment, showing that type-aware auditing and model-specific risk assessment are both necessary for safe agent deployment.

References

Method	F_1	Prec.	Rec.	AUROC
DeBERTa (ours)	**.877**	**.902**	**.854**	**.941**
GPT-4 zero-shot	.710	.680	.740	.780
Keyword baseline	.620	.550	.720	.640

Table 11: Ambiguity classifier performance. DeBERTa substantially outperforms LLM and keyword baselines.

A Mini-Pilot Validation

Before the full study, we conducted a mini-pilot for the two newly-added types (coreferential, authorization scope) using 5 clauses per type $\times$ 2 conditions on GPT-5.4 (20 episodes total). Authorization scope showed 60% violation rate under ambiguity (0% unambiguous); coreferential showed 20% (0% unambiguous). Both patterns were confirmed in the full study (auth-scope: 64% on GPT-5.4; coreferential: 18%), validating the clause design.

B Ambiguity Classifier Proof of Concept

As a preliminary exploration of practical mitigation, we fine-tuned DeBERTa-v3-large as a binary ambiguity detector on 240 training / 60 test clause pairs (Table 11). The classifier achieved $F_1 = 0.877$ (AUROC $= 0.941$), substantially outperforming GPT-4 zero-shot ($F_1 = 0.71$) and keyword baselines ($F_1 = 0.62$). Per-type F_1 showed a layer-consistent pattern: specification-layer types (0.88–0.92) were easier to detect than linguistic-layer types (0.80–0.85), providing additional convergent evidence for the Disambiguation Channel Framework—specification-layer ambiguity involves more salient structural markers, while linguistic-layer ambiguity is more subtle.